

OLISAEMEKA PAUL ANI

opa8@sfu.ca | linkedin.com/in/olisaemeka-paul-ani | github.com/Olisaemeka-Paul-Ani

EDUCATION

Simon Fraser University

B.Sc. Data Science (Computing), Minor in Computing Science, Co-op

Burnaby, BC

Expected Dec 2029

Relevant Coursework: Data Structures & Algorithms, Object-Oriented C++, Memory Management, C++ Templates

EXPERIENCE

Software Engineer

ColorStack SFU

April 2026 – Present

Burnaby, BC

- Deployed a **Python** automation service on **AWS EC2** under systemd with a **Supabase** PostgreSQL backend, delivering job alerts, daily challenges, and a live practice leaderboard to **60 active members** through Slack
- Built a job-scraping pipeline pulling from a GitHub listings repo and the **Bluedoor API**, deduplicating in Supabase and posting new roles 4x daily, surfacing **1,000+ postings** to date for members seeking internships
- Automated daily LeetCode challenges through its **GraphQL** API and a Supabase-backed /leetcode leaderboard
- Traced and resolved a production connection-loss outage with custom application logging built for observability

PROJECTS

AstraFetch | [GitHub](#) | *C++17, libcurl/cpr, std::thread, CMake, vcpkg*

Sep 2026 – Present

- Building a **concurrent** HTTP download manager in C++ that dispatches jobs from a mutex-guarded download queue to a pool of worker threads, tracking each transfer's progress with lock-free std::atomic counters in real time
- Implementing segmented downloads that split files into byte ranges fetched in parallel via HTTP Range requests
- Persisting queue and download state to JSON so interrupted downloads can resume after an application restart

Ferguson | [GitHub](#) | *Go, Bubble Tea, Lip Gloss, REST APIs, Groq, Gemini, Concurrency*

Aug 2026 – Present

- Architected a three-pane FPL analytics terminal in Go around an immutable Model/Update/View state loop, computing pane dimensions from live terminal size to render without content overflow at any window size
- Improved startup responsiveness by dispatching three independent FPL fetches as concurrent commands, cutting data-load latency **62% (835ms to 317ms)** by bounding it to the slowest single call rather than the sum
- Cut player-form API lookups from one per keypress to 15 per session with a lazy, ID-keyed in-memory player cache
- Grounded LLM output in live league data by joining a ~700-player pool against the user's squad to isolate 15 owned players, enriching each with a 5-gameweek fixture difficulty run and serializing into a length-constrained prompt
- Enabled uninterrupted AI verdicts through an automatic Groq-to-Gemini failover mechanism on any provider error

Student Analytics API | [GitHub](#) | *Python, FastAPI, Pydantic, pytest, Ruff*

Jul 2026

- Built a FastAPI **REST** service to log tutoring interactions, with filtered and paginated queries and record deletion
- Computed per-student analytics summaries with trend detection over each student's recorded interaction history
- Enforced request and response contracts with Pydantic v2 schemas, verifying behavior with a six-test pytest suite

LeetLoop | [GitHub](#) | *React Native, Expo, TypeScript, Supabase, OAuth*

Jul 2026 – Aug 2026

- Built a cross-platform React Native and Expo app that keeps students accountable to their daily coding practice
- Developed a custom contribution heatmap from scratch, rendering each user's daily practice as a color-scaled grid
- Designed a five-table Supabase schema for stats screens and user groups, with GitHub and Google **OAuth** sign-in

TECHNICAL SKILLS

Languages: C++, Python, Go, TypeScript, JavaScript, SQL, Bash

Frameworks & Libraries: FastAPI, Pydantic, pytest, Slack Bolt, Bubble Tea, React Native, Expo, cpr/libcurl

Tools & Infrastructure: AWS EC2, Linux, systemd, Supabase, PostgreSQL, Git, GitHub, CMake, vcpkg, Ruff

APIs: REST, GraphQL, OAuth, Claude API, Groq, Gemini